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Chemical Engineering Faculty Retreat Undergraduate Program

Alan Zdunek

UNIVERSITY OF ILLINOIS CHICAGO

UIC Chemical Engineering Undergraduate Program

- EAB & Curriculum Update
- ABET preparation
- Recruitment Best Practices
- 2025-2026 plans

Curriculum Improvement:

CHE 101 Course Revision

Approved by the COE Education Policy Committee (EPC), April 2025

CHE 101: Introduction to Chemical Engineering and Sustainability

Overview of the principles and unit operations of chemical engineering as practiced in traditional industries and in sustainable technologies.

Chemical engineering/unit operations, ChE traditional processes, case studies

Five sustainable themes (modules):
Energy, Health, Food, Air, Water
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Students will explore the challenges in these areas and the role of chemical engineering in developing sustainable products and processes.

Modules will include a guest speaker (faculty or external)

Curriculum Improvement:

Program Revision

ADD CHE 101TO CORE BS REQUIREMENT (+3 CREDIT HOURS)

Initial plan: Remove CHEM 234 Org Chem II

Analysis:

- A review of chemical engineering programs shows two organic chemistry courses are required by Purdue, Northwestern, UW-Madison, UMich-Ann Arbor. UIUC requires Org I, with selective option (Org II or Biochemistry)
- External advisory board: Keep Organic Chem II, important topics covered
- UIC ChE student evaluation: Remove CHEM 234, not interesting, not needed.
- ABET Math/Science course requirements not an issue (minimum 30 credit hours, we have 47 credit hours)

Curriculum Improvement:

Program Revisions

ADDING CHE 101(+3 CREDIT HOURS)

POSSIBLE SOLUTIONS:

- Remove CHEM 234
- Remove CHE 230
- Remove CHEM 342 Physical Chemistry I
- Remove ECE 210 Electric Circuits
- Make CHE 230 and CME 260 a Selective Option
  - Mat Sci requirement of other ChE programs varies: some require it (UMich, NU), some have it as a selective option (UW-Madison), some do not require it in core curriculum (UIUC, Purdue)

CHEM 232 and CHEM 234 Topics

CHEM 232

- Electrons, Bonds, and Molecular Properties
- Molecular Representations
- Acids and Bases
- Alkanes and Cycloalkanes
- Stereoisomerism
- Chemical Reactivity and Mechanisms
- Substitution Reactions
- Alkenes: Structure and Preparation via Elimination Reactions
- Addition Reactions of Alkenes
- Alkynes
- Radical Reactions
- Synthesis
- Alcohols and Phenols
- Ethers and Epoxides; Thiols and Sulfides
- Infrared Spectroscopy and Mass Spectrometry

CHEM 234

- Ethers and Epoxides; Thiols and Sulfides
- Infrared Spectroscopy and Mass Spectrometry
- Nuclear Magnetic Resonance Spectroscopy
- Conjugated Pi Systems and Pericyclic Reactions
- Aromatic Compounds
- Aromatic Substitution Reactions
- Aldehydes and Ketones
- Carboxylic Acids and Their Derivatives
- Alpha Carbon Chemistry: Enols and Enolates
- Amines
- Carbohydrates
- Amino Acids, Peptides, and Proteins
- Lipids
- Synthetic Polymers